

1. Write a procedure that computes the perimeter and the area of a rectangle. Define your own values for the length and width. (Assuming that L and W are the length and width of the rectangle, Perimeter = $2*(L+W)$ and Area = $L*W$).

```
KD2_SAURABH_98912>DELIMITER $$
KD2_SAURABH_98912>
KD2_SAURABH_98912>CREATE PROCEDURE rectangle_values()
-> BEGIN
->   DECLARE L DECIMAL(10,2);
->   DECLARE W DECIMAL(10,2);
->   DECLARE perimeter DECIMAL(10,2);
->   DECLARE area DECIMAL(10,2);
->
->   SET L = 12.00;
->   SET W = 5.00;
->
->   SET perimeter = 2*(L+W);
->   SET area = L*W;
->
->   SELECT area AS AREA_OF_RECTANGLE,
->          perimeter AS PERIMETER_OF_RECTANGLE;
-> END $$
```

Query OK, 0 rows affected (0.01 sec)

```
KD2_SAURABH_98912>
KD2_SAURABH_98912>DELIMITER ;
KD2_SAURABH_98912>call rectangle_values();
```

AREA_OF_RECTANGLE	PERIMETER_OF_RECTANGLE
60.00	34.00

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)

2. Write a procedure that declares an integer variable called num, assigns a value to it, computes and inserts into the temp table the value of the variable itself, its square, and its cube.

```
KD2_SAURABH_98912>
KD2_SAURABH_98912>CREATE PROCEDURE temp_num_powers()
-> BEGIN
->   DECLARE num INT;
->   DECLARE sq INT;
->   DECLARE cb INT;
->
->   SET num = 5;
->   SET sq = num * num;
->   SET cb = num * num * num;
->
->   DROP TEMPORARY TABLE IF EXISTS temp_powers;
->   CREATE TEMPORARY TABLE temp_powers (
->     value_type VARCHAR(20),
->     value INT
->   );
->
->   INSERT INTO temp_powers VALUES ('num', num);
->   INSERT INTO temp_powers VALUES ('square', sq);
->   INSERT INTO temp_powers VALUES ('cube', cb);
->
->   SELECT * FROM temp_powers;
-> END $$
```

Query OK, 0 rows affected (0.01 sec)

```
KD2_SAURABH_98912>
KD2_SAURABH_98912>DELIMITER ;
KD2_SAURABH_98912>CALL temp_num_powers();
```

value_type	value
num	5
square	25
cube	125

3 rows in set (0.03 sec)

Query OK, 0 rows affected (0.03 sec)

3. Create a procedure to Convert a temperature in Fahrenheit (F) to its equivalent in Celsius (C) and vice versa. The required formulae are $C = (F - 32) * 5/9$ $F = 9/5 * C + 32$

```
KD2_SAURABH_98912>CREATE PROCEDURE convert_temp(IN input_value DECIMAL(10,2), IN in_unit CHAR(1))
-> BEGIN
->   DECLARE c DECIMAL(10,2);
->   DECLARE f DECIMAL(10,2);
->
->   IF UPPER(in_unit) = 'F' THEN
->     SET c = (input_value - 32) * 5/9;
->     SET f = input_value;
->     SELECT f AS FAHRENHEIT, c AS CELSIUS;
->   ELSEIF UPPER(in_unit) = 'C' THEN
->     SET c = input_value;
->     SET f = (9/5) * c + 32;
->     SELECT c AS CELSIUS, f AS FAHRENHEIT;
->   ELSE
->     SELECT 'Invalid unit. Use F or C.' AS ERROR_MSG;
->   END IF;
-> END $$
Query OK, 0 rows affected (0.01 sec)

KD2_SAURABH_98912>
KD2_SAURABH_98912>DELIMITER ;
KD2_SAURABH_98912>CALL convert_temp(98, 'F');
+-----+-----+
| FAHRENHEIT | CELSIUS |
+-----+-----+
|      98.00 |    36.67 |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

KD2_SAURABH_98912>CALL convert_temp(37, 'C');
+-----+-----+
| CELSIUS | FAHRENHEIT |
+-----+-----+
|    37.00 |     98.60 |
+-----+-----+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

KD2_SAURABH_98912>
```

4. Create a procedure to Convert a number of inches into yards, feet, and inches. For example, 124 inches equals 3 yards, 1 foot, and 4 inches.

```

KD2_SAUABH_98912>
KD2_SAUABH_98912>CREATE PROCEDURE inches_to_units(IN inches_in INT)
-> BEGIN
->   DECLARE yards INT;
->   DECLARE rem INT;
->   DECLARE feet INT;
->   DECLARE inches_rem INT;
->
->   SET yards = inches_in DIV 36;
->   SET rem = inches_in MOD 36;
->
->   SET feet = rem DIV 12;
->   SET inches_rem = rem MOD 12;
->
->   SELECT yards AS YARDS, feet AS FEET, inches_rem AS INCHES;
-> END $$

```

Query OK, 0 rows affected (0.01 sec)

```

KD2_SAUABH_98912>
KD2_SAUABH_98912>DELIMITER ;
KD2_SAUABH_98912>CALL inches_to_units(124);

```

YARDS	FEET	INCHES
3	1	4

1 row in set (0.00 sec)

Query OK, 0 rows affected (0.00 sec)

5. Create a procedure to read two real numbers and tell whether the product of the two numbers is equal to or greater than 100.

```

KD2_SAURABH_98912>
KD2_SAURABH_98912>CREATE PROCEDURE check_product_100(IN a DECIMAL(10,2), IN b DECIMAL(10,2))
-> BEGIN
->   DECLARE prod DECIMAL(20,4);
->
->   SET prod = a * b;
->
->   IF prod >= 100 THEN
->     SELECT prod AS PRODUCT, 'Product is equal to or greater than 100' AS RESULT;
->   ELSE
->     SELECT prod AS PRODUCT, 'Product is less than 100' AS RESULT;
->   END IF;
-> END $$

```

Query OK, 0 rows affected (0.01 sec)

```

KD2_SAURABH_98912>
KD2_SAURABH_98912>DELIMITER ;
KD2_SAURABH_98912>CALL check_product_100(10, 10);

```

PRODUCT	RESULT
100.0000	Product is equal to or greater than 100

1 row in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)